Tracking the Blast Wave of a Supernova

Radio modelling of the circumstellar environment around SN 2017ein to learn about its progenitor

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ABSTRACT

In this research we attempt to model the circumstellar medium(CSM)/environment around the type Ic supernova SN2017ein using radio data and we compare physical parameters to similar types of supernovae (type Ib, type Ic and radio type II). This is done by applying radio modelled theory including broken power law fits on data received from the Very Large Array assuming the radiation we observe is dominated by synchrotron self absorption (SSA) from an approximated disc in the sky showing the supernova and its blast wave. We calculate radio comparable parameters B = 0.20 ± 0.03 G, R =2.1±0.5×10^16 cm, Ṁ/v_w = 9.4±3.0×10^10 g/cm and ρ_CSM = 1.7±1.0×10−23 g/cm^3. Then by comparing these values to other radio supernovae we conclude our data implies SN2017ein is a relatively low density faint supernova and it is consistent with a binary stripping scenario. Our results are not completely conclusive due to the assumptions and simplifications done in the time frame.

INTRODUCTION

- SN 2017ein is a Type Ic stripped-envelope supernova from the spiral galaxy NGC 3938.
- Supernovae are the explosions caused by the death of massive stars and specifically type Ic are caused from core collapse - the collapse and then “bounce back” of a shock wave of a massive star running out of fuel to fight gravity compressing it. Figure 1 shows an example of the structure.

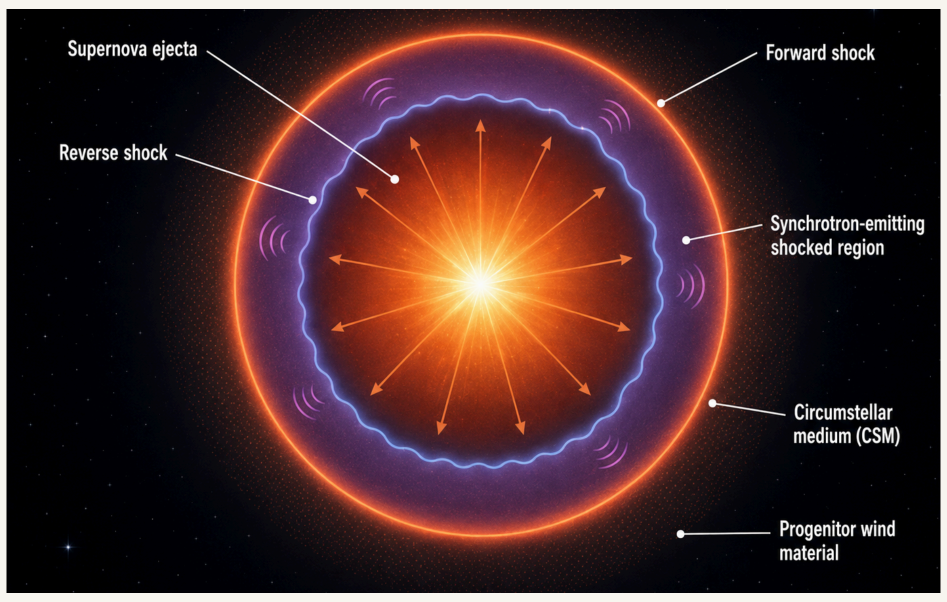

Figure 1: Simple impression of the structure of a supernova. We see the central blast wave pushing material outwards into the CSM and the forward and reverse shock which is what we use to model the CSM. The CSM itself consists of gas and dust which was expelled by the progenitor star from stellar winds or binary interactions. (Binary stripping is when a companion star gravitationally bound attracts and accretes material from the target star through gravity).

- Type Ic spectra lack strong hydrogen and helium features, implying that the progenitor (star that previously existed) lost its outer H/He layers before explosion, consistent with a Wolf-Rayet (WR) star.
- Earlier studies [1] identified a luminous blue pre-explosion source near the SN position, but later HST imaging suggests this source did not disappear and may not be the true progenitor [2], meaning we have not confirmed what the progenitor is.
- Radio emission traces the interaction between the fastest SN ejecta and the surrounding CSM, so if we model it, we can probe the progenitor’s recent mass-loss environment and place constraints on what caused the supernova.
- Aim: correct limiting artifacts and sidelobes from initial images, extract VLA radio brightness from images, fit the radio SED (spectral energy distribution - measure of flux/brightness vs wavelength) with a SSA smooth broken power law, and derive B, R, shock velocity, ρ_CSM and Ṁ/v_w for comparison with other radio supernovae to constrain 2017ein.

METHODS

- Data: VLA radio images of SN 2017ein across L, S, C, X and K bands, covering roughly 1–22 GHz (frequencies known for eq (1)).
- Imaging: CASA tclean in python was used to deconvolve the interferometric dirty image and reduce sidelobe contamination from the dirty beam.
- Host-galaxy contamination: uniform weighting was tested to improve angular resolution and reduce large-scale host emission near the SN position.
- Flux extraction: CASA imfit fitted a 2D Gaussian at the SN position. The lower-frequency L-band point and K-band point were treated cautiously as non-detections/upper limits.
- SED fitting: the measured fluxes were fitted (figure 2) with a smooth broken power law from eq (1), with constraints for s, α_1 and optically thin slope α_2 fitted (least squares used in our own jupyter notebooks).
- Physical modelling: F_brk and ν_brk were converted into B, R, ρ_CSM and Ṁ/v_w using the SSA formalism and scaling relations in python used by DeMarchi et al 2022 [3].

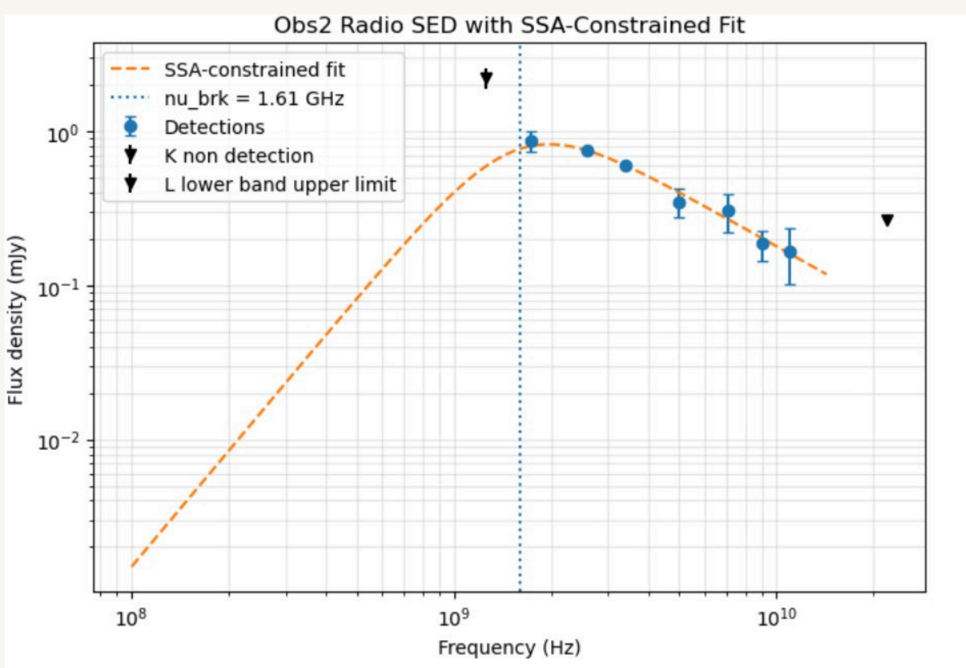

Figure 2: SED fit for our flux measurements from the radio images. The black arrows correspond to non detections and the orange line shows the model fitted using eq (1). We see one slope corresponding to α_1 and the other corresponding to α_2 . The fitted turnover values are then used to compute physical properties of the CSM.

RADIO MODELLING

- The radio SED (Figure 2) was fitted using a synchrotron self-absorbed smooth broken power law:

$$F_\nu = F_{\rm brk}\left[\left(\frac{\nu}{\nu_{\rm brk}}\right)^{\alpha_1/s} + \left(\frac{\nu}{\nu_{\rm brk}}\right)^{\alpha_2/s}\right]^s \qquad (1)$$

- This power law is derived from assuming that synchrotron emission dominates the data that we observe which is caused by the forward shock wave accelerating electrons in the CSM.
- The SED turnover is described by a break frequency, ν_brk, and break flux, F_brk. These fitted break values are then used in the SSA scaling relations to derive physical properties of the shock and CSM [3].
- Constraints used in the fit: s = −1 [3], optically thick slope $\alpha_1 = 5/2$, and $-3 < \alpha_2 < 0$ [4].
- Best fit: Fbrk = 1.54 ± 0.07 mJy, νbrk = 1.6 ± 0.2 GHz, $\alpha_2 = -1.2 \pm 0.2$.
- Physical parameters were then derived from the fitted break values, which follows the radio modelling in [3] and [5]:

$$B = 0.5761\left(\frac{\nu_{\rm brk}}{5\,\rm GHz}\right)\left[\epsilon_B^{-1}\left(\frac{D}{\rm Mpc}\right)\epsilon_e\left(\frac{F_{\rm brk}}{\rm Jy}\right)^{1/2}\right]^{-4/19}\ {\rm G} \qquad (2)$$

$$R = 8.759\times10^{15}\left(\frac{\nu_{\rm brk}}{5\,\rm GHz}\right)^{-1}\left[\left(\frac{D}{\rm Mpc}\right)^{18}\epsilon_B\left(\frac{F_{\rm brk}}{\rm Jy}\right)^{9}\epsilon_e^{-1}\right]^{1/19}\ {\rm cm} \qquad (3)$$

- Using t = 12 ± 0.3 days [6] and D = 19.1 ± 3.5 Mpc [7], equipartition (ϵB = ϵe = 0.33) gives:
- B = 0.20 ± 0.03 G,   R = $(2.1 \pm 0.5)\times10^{16}$ cm
- Wind-density relation: ρCSM = $\dot{M}/(4\pi R^2 v_w)$ and Ṁ/v_w can be calculated from the same modelling used to find B and R [3], [5].
- Final values: ρCSM = $(1.7 \pm 1.0)\times10^{-23}$ g cm^{-3} and Ṁ/vw = $(9.4 \pm 3.0)\times10^{10}$ g cm^{-1}.

KEY RESULTS

Magnetic Field · equipartition
B = 0.20 ± 0.03 G

Shock Radius · from SSA scaling
R = $(2.1 \pm 0.5)\times10^{16}$ cm

Mass-loss Parameter
Ṁ/v_w = $(9.4 \pm 3.0)\times10^{10}$ g/cm

CSM Density · equipartition
ρ = $(1.7 \pm 1.0)\times10^{-23}$ g/cm³

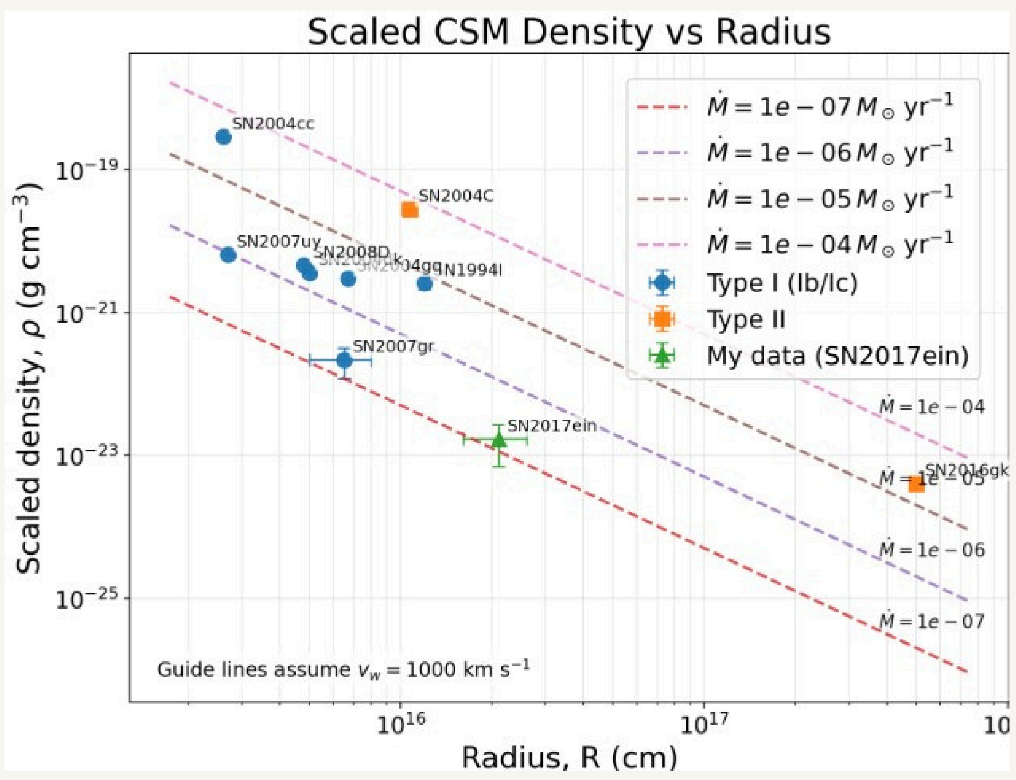

Figure 3: Density scaled to our equipartition values vs radius/radial distance from the initial explosion for each supernova we identified for comparison analysis. Type I and Type II are split up. Scaling lines show how different densities fit on different mass loss rates. SN2017 is separately identified by the green point.

IMPLICATIONS

- From figure 3, we can see the derived CSM density is low and sits on a low mass loss rate compared with many similar radio-detected stripped-envelope SNe, suggesting a relatively tenuous environment around SN 2017ein.
- The mass-loss parameter Ṁ/v_w depends strongly on the assumed wind speed and microphysical parameters, so it should not be overinterpreted as a unique progenitor mass-loss rate.
- Non-detections and possible host-galaxy contamination remain limitations in the fit.
- The low CSM density and mass loss rate grouping is consistent with a binary stripped-envelope progenitor in a tenuous wind environment [8].
- Our radio data alone cannot uniquely determine the progenitor channel.

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